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Clinical Investigation: Central Nervous System Tumor

Outcomes of Proton Radiation Therapy for Peripapillary Choroidal Melanoma at the BC Cancer Agency

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Summary

We report our experience with proton beam therapy in the treatment of peripapillary choroidal melanomas by means of a retrospective analysis of 59 patients treated between 1995 and 2007 at the only Canadian proton therapy facility. This article shows that excellent control and toxicity rates can also be achieved at smaller proton centers, which is of importance as newer centers come online and may provide this treatment option for localized choroidal melanomas. These results also raise caution for larger (T3) peripapillary tumors, because this group appears to have worse local control and distant metastasis rates.

Purpose: To report toxicity, local control, enucleation, and survival rates for patients with peripapillary choroidal melanoma treated with proton therapy in Canada.

Methods and Materials: We performed a retrospective analysis of patients with peripapillary choroidal melanoma ($\leq$2 mm from optic disc) treated between 1995 and 2007 at the only Canadian proton therapy facility. A prospective database was updated for follow-up information from a chart review. Descriptive and actuarial data are presented.

Results: In total, 59 patients were treated. The median age was 59 years. According to the 2010 American Joint Committee on Cancer TNM classification, there were 20 T1 tumors (34%), 28 T2 tumors (48%), and 11 T3 tumors (19%). The median tumor diameter was 11.4 mm, and the median thickness was 3.5 mm. Median follow-up was 63 months. Nineteen patients received 54 cobalt gray equivalents (CGE) and forty patients received 60 CGE, each in 4 fractions. The 5-year actuarial local control rate was 91% (T1, 100%; T2, 93%; and T3, 59%) ($p = 0.038$). There was a suggestive relationship between local control and dose. The local control rate was 97% with 60 CGE and 83% with 54 CGE ($p = 0.106$). The metastasis-free survival rate was 82% and related to T stage (T1, 94%; T2, 84%; and T3, 47%) ($p < 0.001$). Twelve patients died, including eleven with metastases. The 5-year actuarial rate of neovascular glaucoma was 31% (23% for T1−T2 and 68% for T3, $p < 0.001$), and that of enucleation was 0% for T1, 14% for T2, and 72% for T3 ($p < 0.001$). Radiation retinopathy (74%) and optic neuropathy (64%) were common within-field effects.

Conclusions: Proton therapy provides excellent local control with acceptable toxicity while conserving the globe in 80% of cases. These results are consistent with other single-institution series using proton radiotherapy, and toxicity rates were acceptable. T3 tumors carry a higher rate of both local recurrence and metastasis. © 2012 Elsevier Inc.

Keywords: Proton therapy, Choroidal melanoma, Peripapillary, Local control, Survival

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